

Review

Insect adaptation to urban environments: a synthesis of morphological, behavioural and physiological responses

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Urbanisation is among the most pervasive drivers of contemporary environmental change, imposing a complex suite of abiotic and biotic pressures on insect populations worldwide. Here, we synthesise recent evidence on how insects respond to urban environments across three interconnected axes: morphology, behaviour and physiology. Urban heat and dry islands, habitat fragmentation, artificial light at night, noise, air and chemical pollution and novel plant communities act as multidimensional filters that can drive shifts in body size, locomotory appendage, colouration, foraging and mating behaviours, activity patterns, thermal physiology, water balance, stress metabolism and seasonality. These changes in morphology, behaviour and physiology do not act independently but instead form an integrated functional response to urban environments. Moreover, multiple urban stressors, sometimes acting in opposing directions, can interact to shape trait responses. We further discuss the concept of urban trait syndromes, wherein suites of correlated traits co-vary or are co-selected under shared urban pressures. Insect responses to urbanisation arise through both phenotypic plasticity and evolutionary adaptation, though disentangling their relative contributions remains a key challenge. In some cases, maladaptive responses can arise when historically reliable cues become decoupled from fitness in novel environments. Finally, we identify critical knowledge gaps, including the need for more multi-stressor experiments, greater taxonomic and geographic coverage and long-term studies that track the pace and limits of urban adaptation.

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Introduction

Urbanisation ranks among the most rapid and pervasive drivers of global environmental change, converting vast portions of the Earth's surface into human-dominated ecosystems that depart markedly from the evolutionary history of most organisms. For insects, which dominate terrestrial biodiversity and underpin critical ecosystem services such as pollination, decomposition and pest regulation, cities represent complex selective landscapes where multiple novel abiotic and biotic pressures act as rigorous multidimensional filters [1]. This environmental sieve excludes species lacking suitable pre-adaptations while exerting strong directional selection on those that persist [2], frequently favouring a suite of traits including accelerated development, reduced body size, increased boldness, broader resource use and greater thermal tolerance.

The urban abiotic environment is primarily characterised by increases in built-up structures and impervious surfaces, creating urban heat islands (UHIs) [3] with heterogeneous microclimates that challenge organisms' thermal performance and the seasonal timing of life-history events. In addition to thermal modifications, impervious surfaces can restructure hydrological regimes, producing urban dry island effects [4] and localising moisture accumulation [5]. Urban built-up structures alter near-surface airflow dynamics and reduce average wind speeds, while large buildings create localised wind tunnels with strong turbulence that can disrupt semiochemical communication and constrain flight efficiency [6]. Habitat fragmentation isolates remnant green spaces, reducing dispersal and gene flow [7], whereas anthropogenic stressors such as artificial light at night (ALAN), noise and chemical pollutants, disrupt circadian regulation and physiological homeostasis and often act synergistically with thermal stress. The replacement of native vegetation with non-native ornamental species restructures plant communities, disrupting host-plant recognition and phenological synchrony between interacting species [8,9].

Critically, these stressors rarely act in isolation. In complex urban ecosystems, organisms experience multiple interacting pressures that often generate synergistic effects [2]: elevated temperatures amplify pesticide toxicity [10], while homogenised green spaces reduce floral resource quality and impair pollinator immune function [11]. Understanding whether and how insects adjust to these multifaceted challenges — through immediate

plastic responses, longer-term evolutionary changes, or a combination of both — is essential for predicting urban biodiversity trajectories and sustaining ecosystem services in cities where nearly 70% of the global population is projected to reside by 2050 [12].

Morphological responses

Body size is the most extensively studied morphological trait in urban ecological and evolutionary entomology, and the accumulated evidence reveals a nuanced picture shaped by competing selective pressures. On the one hand, the temperature–size rule predicts that UHI-driven warming will reduce body size by accelerating development [13]; on the other hand, anthropogenic subsidies, such as garden plants, can support larger body sizes by enhancing resource availability [14]. These opposing forces generate idiosyncratic outcomes across taxa. Smaller body sizes have been documented in ground beetles along rural–urban gradients [15] and in bumblebees as temperature increases across urbanisation gradients [16]. Conversely, common garden experiments demonstrate that urban-origin grasshoppers grow faster and attain a larger body mass under elevated temperatures [17], and some wild bee species have larger body sizes at medium-to-high urbanisation levels, potentially reflecting abundant urban floral resources [18]. Macromoths also exhibit increased body sizes along urbanisation gradients at both intraspecific and community levels, which may be driven by selection for mobile individuals and species in fragmented habitats [19]. A global synthesis further reveals heterogeneous responses of body size to urbanisation among insect taxa, with Coleoptera shifting towards smaller and Lepidoptera shifting towards larger body sizes [20]. This variation underscores that insect body size responses are species-specific and reflect the net effect of various filtering mechanisms such as thermal conditions, habitat fragmentation and resource availability, which can operate at different levels of biological organisation (e.g. individual vs. community) [21].

Another morphological trait closely related to body size is locomotory appendage (e.g. wings and legs), which responds to a different set of urban pressures centred on movement and dispersal. Habitat fragmentation generally favours more mobile morphs, and this prediction is broadly supported: urban grasshoppers develop longer wings and femurs [17], and flightless ground beetles exhibit enlarged hind tibiae that may confer locomotory advantages in patchy urban landscapes [22]. Yet, this trend is not universal, and other factors can impose a countervailing selective pressure. For example, ALAN renders flight costly through disorientation and increased predation mortality around light sources. Consequently, urban spindle ermine moths have evolved smaller wings [23]. Colouration represents a further axis of

morphological change. Urban pollutants lead to industrial melanism of insects because dark colouration confers selected advantages under predation, as seen in the peppered moth [24]. On the other hand, warmer urban microclimates can favour lighter pigmentation in social wasps [25], consistent with the thermal melanism hypothesis [26]. Together, these morphological shifts illustrate how urban environments can simultaneously select for altered mobility and thermal regulation, sometimes in opposing directions within the same landscape, although more research, especially on less-studied taxa, is needed to better elucidate those mechanisms.

Behavioural responses

Urban environments alter the sensory world of insects in ways that cascade through foraging, mating and broader behavioural repertoires [27]. A central mechanism involves the disruption of volatile organic compound (VOC) signalling. Elevated concentrations of ozone and other pollutants react with plant-emitted VOCs, creating altered ‘odorscapes’ that impair host-finding across multiple steps, from emission to transmission and perception [28]. The consequences can be profound: ozone-treated floral blends become less attractive to hawk-moths [29], and diesel-exposed honeybees show impaired olfactory learning [30]. Furthermore, chemical pollutants may influence foraging indirectly via changes in food resources. For example, elevated CO₂ can modify host plant quality and defences, prompting insects to alter their consumption to meet nutritional needs [31].

Mate finding and courtship are similarly vulnerable because those processes depend on visual, chemical and acoustic cues that are impacted by various types of urban pollution [32]. ALAN interferes with mate signalling in fireflies [33] and activity levels of glow-worm larvae [34]. Ozone exposure delays female acceptance and increases male–male courtship in fruit flies [35]. Anthropogenic noise reduces phonotaxis in field crickets [36] while shifting call frequencies in roadside grasshopper populations [37]. The common thread is that urban sensory pollution erodes the reliability of communication channels, forcing behavioural adjustments that may carry fitness costs. Another disrupted reproductive behaviour is oviposition site selection: female butterflies oviposit on non-native host plants that cannot support larval development, creating evolutionary traps in which historically reliable olfactory cues now lead to maladaptive outcomes [38].

Activity patterns shift in response to both ALAN and UHIs. Dim light at night increases nocturnal activity in a diurnal fruit fly while reducing its total daily activity [39]. ALAN also increases nocturnal biting in a diurnal

disease-vector mosquito [40]. UHIs boost foraging activity in ants [41] and stingless bees [42]. Furthermore, urban populations tend to display increased boldness and exploratory behaviours — documented in ground beetles [43] and grasshoppers [44] — although such personality shifts are species-specific and not universal [45]. Finally, perhaps most consequential for survival, ALAN can alter anti-predator behaviours in moths by suppressing ultrasound-evoked evasive flights, thereby increasing vulnerability to bat predation [46].

Physiological responses

Thermal physiology represents one of the most well-studied physiological axes along which urban insect populations diverge from their rural counterparts. A common garden study on fruit flies revealed that urban populations exhibit higher heat tolerance but reduced cold tolerance [47], consistent with directional selection imposed by elevated urban temperatures. Similarly, common-garden-reared adult moths from urban populations show higher heat shock tolerance than their rural counterparts [48]. On the other hand, asymmetric responses at two ends of an organism's thermal range — reduced cold tolerance yet without a corresponding increase in heat tolerance — has been reported in urban ants [41] and bees [49], leading to a narrower thermal breadth. Besides thermal limits, thermal optima can shift as well: urban ants show increased thermal optima for locomotor performance that enable more efficient resource retrieval under warm conditions [50]. It is worth noting that most empirical evidence comes from temperate cities, while studies are currently lacking in warmer climates, where insect thermal responses to urbanisation can be quite different.

Water balance represents an underappreciated physiological challenge in cities, where UHIs increase the vapour pressure deficit. Although research on the desiccation tolerance of urban insects remains limited [51], emerging evidence suggests that dietary shifts can serve as a coping mechanism: urban ants show increased lipid consumption relative to protein, producing more metabolic water that helps ease desiccation stress [52]. Urban stressors also provoke broader metabolic, endocrine and immune responses. For example, honeybee digestive enzyme (aminopeptidase N) activity declines at warmer urban sites, indicating higher stress levels [53]. Heavy metal pollution disrupts hormone signalling in midges [54] and suppresses immunity in ground beetles [55]. Physiological responses are also tightly linked to life-history patterns: higher urban temperatures generally favour 'fast' strategies characterised by accelerated growth and earlier maturation, as documented in mosquito larvae [56]. However, reduced growth and development may also occur when urban temperatures exceed the thermal optima of organisms

[57] or other factors, such as predation regimes override thermal effects [58], underlining the context dependence of life-history shifts in cities.

Urbanisation influences the seasonal physiology of insects by altering fall diapause and spring emergence [59]. Diapause is primarily regulated by photoperiod, which can be disrupted by light pollution: ALAN suppresses diapause in moths [60], butterflies [61] and mosquitoes [62]. In the case of mosquitoes, diapause suppression is often associated with increased blood feeding and reproductive activities, potentially facilitating disease transmission. Spring emergence is largely governed by temperature, and urban warming can lead to advanced emergence, as has been documented in butterflies [63] and mosquitoes [64]. Interestingly, besides ALAN and UHIs, abundant urban floral resources can also affect insect seasonality by broadening the activity periods of pollinators [65]. Together, these urbanisation-mediated shifts in seasonal patterns can have profound consequences for population dynamics through changes in overwintering survival and voltinism.

Interplay between responses and urban trait syndromes

The morphological, behavioural and physiological responses reviewed above do not act independently but instead form an integrated functional response to urban environments. For instance, increased metabolic rates due to UHI effects may drive bolder foraging behaviour to meet elevated energy demands. Behavioural thermoregulation can buffer physiological stress and thereby reduce the intensity of selection on thermal tolerance — a phenomenon termed the 'Bogert effect' [66], highlighting that the observed changes in thermal physiology in response to urbanisation likely reflect the selection pressure that remains after behavioural compensation. Furthermore, multiple urban stressors can interact to shape insect responses, both synergistically and antagonistically [67]. Current research has mostly focused on the interactions between ALAN and UHI effects. For instance, ALAN and urban warming have synergistic effects on the nighttime activities of mosquitoes [68] and the suppression of diapause in flesh flies [69] while having antagonistic effects on the development of female field crickets [70].

Suites of traits can co-vary or be co-selected under shared urban pressures, giving rise to 'urban trait syndromes' [71]: smaller body sizes, bolder behaviours, higher activity levels, greater dietary breadth, faster development and increased thermal tolerance. A global meta-analysis of over 300 ground beetle species revealed that urbanisation selects for higher mobility and generalist feeding strategies, generating trait syndromes that enhance resource exploitation in fragmented, disturbed

Table 1**Evidence of insect morphological, behavioural and physiological responses to urbanisation.**

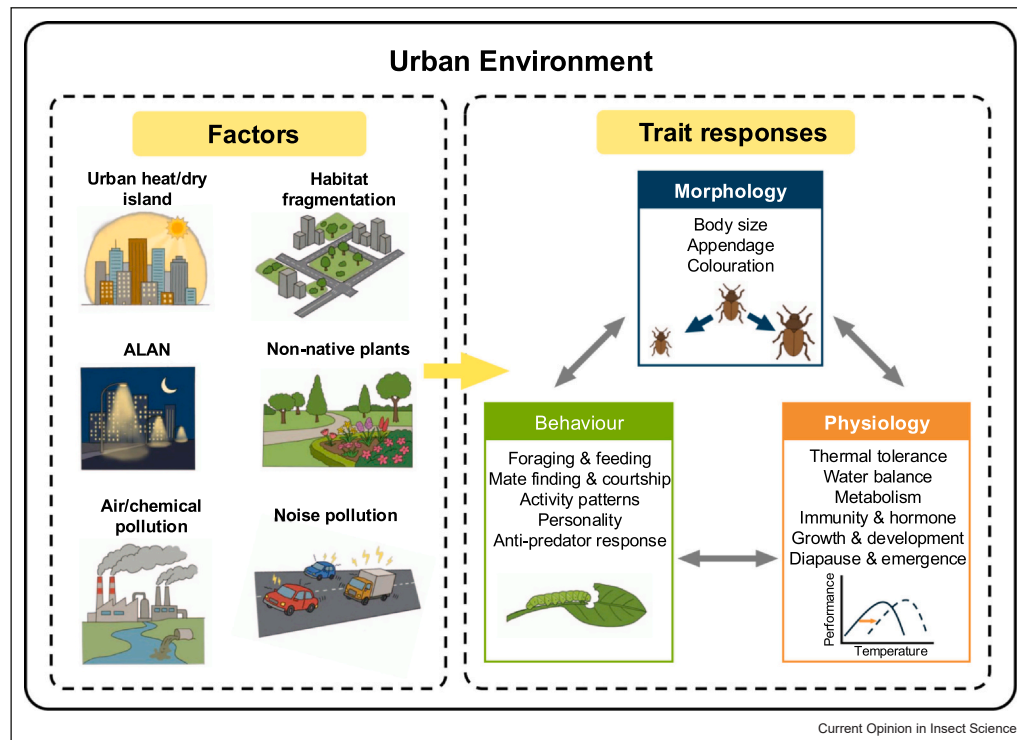
Category	Trait	Response	Factor	Example
Morphology	Body size	Reduced body size	UHI effects	Ground beetle [15], bumblebee [16]
		Increased body size	Abundant floral resources	Wild bee [14,18]
	Locomotory appendage	Increased body size	Habitat fragmentation	Moth [19]
		Increased wing size	Habitat fragmentation	Grasshopper [17]
		Reduced wing loading	ALAN	Moth [23]
		Increased wing fluctuating asymmetry	Habitat fragmentation	Hole-nesting bee [77]
		Increased femur length	Habitat fragmentation	Ground-nesting bee [77], wasp [77], bumblebee [16]
		Increased hind tibia size	Habitat fragmentation	Grasshopper [17]
	Colouration	Increased melanism	Air pollution	Ground beetle [22]
		Increased brightness	UHI effects	Moth [24]
Behaviour	Foraging	Disrupted host finding	UHI effects	Wasp [25]
	Reproduction	Altered mate signalling	Air pollution	Moth [29], honeybee [30]
		Altered courtship behaviour	ALAN	Firefly [33]
	Activity pattern	Reduced phonotaxis	Air pollution	Fruit fly [35]
		Altered mate signalling	Noise pollution	Crickets [36]
		Altered oviposition site selection	Noise pollution	Grasshopper [37]
		Increased nighttime activity	Non-native host plants	Butterfly [38]
		Reduced nighttime activity	ALAN	Fruit fly [39], mosquito [40]
		Increased daytime activity	ALAN	Glow-worm [34]
Physiology	Personality	Increased boldness and risk-taking	UHI effects	Ant [41], stingless bee [42]
		Suppressed evasive flight	Anthropogenetic stressors	Ground beetle [43], grasshopper [44]
	Anti-predator behaviour	Increased heat tolerance	ALAN	Moth [46]
	Thermal tolerance	Reduced cold tolerance	UHI effects	Fruit fly [47], moth [48]
		Increased thermal optimum	UHI effects	Fruit fly [47], ant [41], honeybee [49]
	Osmoregulation	Increased lipid consumption for metabolic water	UHI effects and dessication stress	Ant [50]
	Stress	Increased stress level	UHI effects	Ant [52]
	Body function	Disrupted hormone signalling	UHI effects	Honeybee [53]
		Disrupted immunity	Chemical (heavy metal) pollution	Midge [54]
	Growth and development	Increased growth and development	Chemical (heavy metal) pollution	Ground beetle [55]

environments [71]. Understanding how traits co-vary in response to urbanisation remains a key knowledge gap. Furthermore, traits can potentially become desynchronised when they respond to different primary factors, and such decoupling could generate novel trade-offs that constrain the long-term adaptive potential of urban insect populations. Finally, different urban stressors can lead to opposing trait responses. For example, UHIs favour smaller body sizes, whereas habitat fragmentation favours larger body sizes; UHIs favour lighter colouration, whereas air pollution favours darker colouration. The net outcome will depend on the relative strength of selection by each stressor.

Mechanisms of adaptation: plasticity vs. evolution

The responses documented above arise through two fundamentally different mechanisms that operate across different timescales [72]. Phenotypic plasticity provides immediate buffering, allowing a single genotype to produce multiple phenotypes in response to environmental variation [73]. Evolutionary adaptation, by contrast, involves heritable allele-frequency shifts that provide a basis for longer-term population persistence. Disentangling these two mechanisms is a key challenge in urban evolutionary ecology because both can produce similar phenotypic outcomes. Common garden

Figure 1



Urban insect populations encounter multidimensional filters and interacting stressors that shape responses across morphological, behavioural and physiological axes, frequently generating urban trait syndromes. These integrated responses can facilitate short- and long-term population persistence through phenotypic plasticity, which provides immediate buffering, and through directional selection, which drives evolutionary adaptation.

experiments offer the most direct approach to this. For example, increased thermal tolerance in urban cabbage white butterflies has been attributed to plasticity rather than genetic changes [74]. Interestingly, sympatric species sharing the same urban habitat can adopt contrasting strategies. For example, urban Chinese crickets increase their body size under predation pressure, likely through evolutionary changes, whereas co-occurring stove grasshoppers decrease their body size under energetic constraints, likely through phenotypic plasticity [75]. Such divergent responses highlight how species-specific responses can channel adaptation along distinct pathways even under shared environmental conditions. We refer the readers to Daimond et al. [76] for a detailed review of insect adaptations to urban environments.

When environmental change is extreme or rapid, developmental stability can break down through decanalisation, a process that exposes previously buffered phenotypic variation to selection. Increased fluctuating asymmetry in wing morphology, which can have profound impacts on flight performance, has been documented in urban bees and wasps [16,77], providing evidence that intensifying urban stressors can exceed the capacity of developmental buffering systems. At the

opposite end of the adaptive spectrum, maladaptation through ecological and evolutionary traps represents a critical risk when historically reliable environmental cues become decoupled from fitness outcomes. Polarised light reflected from artificial surfaces such as solar panels or sports fields attracts aquatic insects to unsuitable oviposition sites [78]; caddisfly larvae incorporate anthropogenic materials into protective cases with reduced structural integrity [79]. However, maladaptation is not necessarily a terminal state. Urban spindle ermine moths that were initially attracted to artificial light have subsequently evolved reduced light attraction accompanied by smaller wings and reduced flight capacity [23], illustrating how maladaptation can transition to adaptation over time given sufficient genetic variation in the population.

Conclusions and future directions

Research over the past decade has substantially advanced our understanding of how insects respond to urbanisation, revealing widespread morphological, behavioural and physiological adjustments that are linked by common selective pressures (Table 1; Figure 1). Several general patterns emerge: urban populations frequently exhibit altered body size, enhanced mobility,

modified foraging and mating behaviour and shifted thermal physiology. Although many insects are negatively impacted by urban stressors, some may actually benefit from them. For instance, the reproduction of a scale insect increases with urban warming and drought, leading to higher fitness and abundance [80]. Therefore, there will likely be urban ‘winners’ and ‘losers’, and understanding the relationship between trait performance and various stressors will help predict how insect communities may shift under urbanisation. Furthermore, trait responses can be highly species- and taxon-specific, yet current research is largely limited to a few insect groups (e.g. ground beetles, moths, bees, fruit flies and crickets), highlighting the need for broader taxonomic coverage. Finally, cities in different geographic regions vary in their climatic conditions, physical environments and human activities, all of which can modulate insect responses to urbanisation. However, most studies have been conducted in temperate regions in North America and Europe, stressing the need for larger spatial coverage across cities in the tropics and on other continents. Addressing the taxonomic and geographic disparities will allow for a more complete picture of insect responses to urbanisation.

Several key priorities for future research include: (1) multi-stressor experiments that examine how urbanisation factors such as UHI effects, ALAN, air pollution and habitat fragmentation interact to shape insect responses; (2) common garden and reciprocal transplant experiments that disentangle the relative contributions of phenotypic plasticity and evolutionary adaptation across different trait axes; (3) more attention on understudied trophic guilds such as detritivores and scavengers, whose ecosystem functions are critical but responses to urbanisation remain poorly characterised; (4) investigations into how different trait responses interact with or constrain each other to shape insect performance and fitness; and (5) long-term monitoring that tracks the pace and limits of urban adaptation across generations. Understanding the mechanisms and the constraints underlying insect adjustments to urban environments is essential for conserving biodiversity and sustaining ecosystem services in an increasingly urbanised world.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Syuan-Jyun Sun reports financial support was provided by National Science and Technology Council. Syuan-Jyun Sun reports financial

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